PRODUCT REQUIREMENTS DOCUMENT

EduPath Consultancy

Education Consultancy Management System – college admissions, counselling and enquiry management for Indian students

Document version	1.0
Date	31 October 2026
Product status	Built and tested in development; pre-launch
Platform	Responsive web application (desktop and mobile browsers)
Technology	Next.js 16 · React 19 · Express 5 · Node.js · MongoDB (Mongoose 8) · TypeScript
Owner	ramsah01012002@gmail.com

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1. Overview and problem statement

Students in India choosing a college face scattered information: course lists, eligibility rules, fees and deadlines are spread across hundreds of college websites, and fees are often out of date. Education consultancies that help these students usually track enquiries in spreadsheets and phone notes, which makes follow-ups slip, hides counsellor workload, and gives management no reliable reports.

EduPath Consultancy is a single web platform where:

- **Students** discover colleges and courses, check eligibility, compare verified fees, submit admission enquiries and track their progress.
- **Counsellors** work only on the enquiries assigned to them: call, WhatsApp, record notes, check eligibility, recommend courses and move each enquiry through the admission pipeline.
- **Admins** manage the catalogue, fees, counsellors, enquiries, scholarships, content and reports, each action protected by fine-grained permissions and recorded in an audit log.

2. Goals and non-goals

Goals

- Turn every student enquiry into a tracked case with a reference number, an owner and a clear status.
- Give students honest, source-backed college information: no fabricated fees, images or rankings.
- Let counsellors give eligibility-based advice quickly using the student's actual marks.
- Give admins live visibility into pipeline, counsellor performance and conversion.
- Protect student personal data: strict access isolation, verified emails, private file storage and data-correction/deletion requests.

Non-goals (this release)

- Native Android/iOS apps (the web app is mobile-responsive instead).
- Applying to colleges on the student's behalf through college portals.
- Collecting college tuition fees; online payment covers only the consultancy's own service charge.
- Guaranteed admission or guaranteed eligibility – the system only ever says "eligible", "potentially eligible" or "not eligible" with reasons.

3. Users and roles

Role	Who	How the account is created	What they can access
Student	A prospective student or parent	Public sign-up (default role), email OTP verification	Only their own profile, enquiries, applications, appointments, files, feedback and payments
Counsellor	Consultancy staff who advise students	Sign-up with a single-use invitation code, or created by an admin; email OTP verification	Only enquiries assigned to them (any other returns 404). No access to `/api/admin`.
Admin	Consultancy management and operations	Invitation code, admin invitation, or the one-time `admin:bootstrap` command; optional two-step sign-in (authenticator app)	Admin area, limited by the permissions granted

Admin permissions

Permissions are granted per admin so each person gets only what they need:

Permission	Allows	Permission	Allows
`students.read`	View students, dashboard	`scholarships.manage`	Scholarship rules
`students.write`	Correct student details	`scholarships.approve`	Scholarship decisions
`admins.manage`	Admins, counsellors, invite codes	`reports.export`	Reports and CSV export
`catalog.manage`	Colleges, courses, media	`feedback.manage`	Feedback and testimonials
`pricing.manage`	Fees and fee publishing	`privacy.manage`	Data correction/deletion requests
`enquiries.manage`	Enquiry CRM, assignment	`content.manage`	Pages and announcements
`admissions.confirm`	Mark an enquiry ADMITTED	`payments.manage`	Payment requests
`email.view`	Email outbox	`settings.manage`	Site settings, email templates
`audit.read`	Audit log		

4. Key user journeys

4.1 Student: from search to admission

```
Sign up → verify email (6-digit OTP) → complete academic profile (Class 10 / 12 / graduation marks)
→ search colleges (name, state, city, branch) or browse courses → check eligibility & compare fees
→ submit enquiry → receive reference number (ENQ-YYYY-…) + acknowledgement email
→ talk to assigned counsellor → see recommendations, upload requested documents, attend counselling
→ track status until ADMITTED
```

4.2 Counsellor: working an enquiry

```
Sign in → "My enquiries" (assigned only, with follow-up dates) → open enquiry
→ Call / WhatsApp / copy phone → log contact attempt & notes → run eligibility check against an offering
→ recommend courses (visible to student) → move status (e.g. ELIGIBILITY_CHECK → COUNSELLING_COMPLETED)
→ daily follow-up digest email at 09:00 IST
```

4.3 Admin: running the consultancy

```
Sign in (optional 2-step) → dashboard totals & charts → assign enquiries (or enable auto-assign)
→ manage colleges, courses, fees, media → create counsellors / invitation codes
→ review counsellor performance & enquiry reports → export CSV → confirm admissions
```

4.4 Enquiry status pipeline

Main pipeline: NEW → CONTACT_PENDING → CONTACTED → INTERESTED → ELIGIBILITY_CHECK → COUNSELLING_SCHEDULED → COUNSELLING_COMPLETED → DOCUMENTS_PENDING → APPLICATION_IN_PROGRESS → SUBMITTED → OFFER_RECEIVED → ADMITTED

Side statuses: ON_HOLD, NOT_INTERESTED, REJECTED, WITHDRAWN, CLOSED.

- Forward moves and one step back are allowed; pausing, ending, reopening and going back require a reason.
- Evidence is required before SUBMITTED and OFFER_RECEIVED. ADMITTED needs an admin with `admissions.confirm`.
- Some moves are automatic: a reached contact sets CONTACTED, the first booked session sets COUNSELLING_SCHEDULED, a document request sets DOCUMENTS_PENDING.
- Every change is recorded in an append-only history; important changes email the student.

5. Functional requirements

Priority: **P0** = required for launch, **P1** = important, **P2** = nice to have. Status reflects the build as of 31 Oct 2026.

5.1 Accounts and authentication

ID	Requirement	Priority	Status
AUTH-1	Sign-up with name, unique email, mobile and strong password; "I am a" choice of Student, Counsellor or Admin	P0	Done
AUTH-2	Counsellor/Admin sign-up requires a valid, single-use, expiring invitation code; the role comes from the code, never the request	P0	Done
AUTH-3	Email verification by 6-digit OTP for every role before first login; resend with rate limit	P0	Done
AUTH-4	Reject email domains that cannot receive mail (MX/A check)	P1	Done
AUTH-5	Password reset by email link; passwords hashed with argon2	P0	Done
AUTH-6	Server-side sessions in HttpOnly cookies, expiry, logout; role-based home page	P0	Done
AUTH-7	Optional two-step sign-in for admins (authenticator app + 8 recovery codes)	P1	Done
AUTH-8	Admins create, list and revoke invitation codes; CLI `npm run invite:create` for first setup	P0	Done

5.2 Student profile and dashboard

ID	Requirement	Priority	Status
STU-1	Academic profile: Class 10, Class 12 (with stream) and graduation marks, entrance exams, preferences; percentage derived on the server	P0	Done
STU-2	Dashboard: recommendations with eligibility badges, shortlist, enquiries, assigned counsellor (name only), notifications, announcements	P0	Done
STU-3	Profile photo upload (JPG/PNG/WebP, 2 MB, content-checked)	P2	Done
STU-4	Shortlist colleges and save fee comparisons	P1	Done
STU-5	Request correction or deletion of personal data	P1	Done
STU-6	Contact the consultancy on WhatsApp and see its social links	P1	Done

5.3 College and course discovery

ID	Requirement	Priority	Status
CAT-1	Real-time college finder: search as you type by name, state, city and branch (engineering, medical, management, law, etc.); common abbreviations (IIT, NIT, AIIMS, DU…); link to each college's official website. Data: ~26,000 Indian institutions from Wikidata (CC0), refreshed weekly	P0	Done
CAT-2	Verified directory of published colleges with filters, sorting, detail page, gallery, facilities, exams, deadlines	P0	Done
CAT-3	"India's best colleges" cards and large campus photo showcase (NIRF 2024 top 10, licensed Wikimedia Commons photos with credits)	P2	Done
CAT-4	Browse courses across colleges: filter by level, state/city, maximum fee, qualification, mode (online/offline/hybrid); sort by deadline or fee	P1	Done
CAT-5	College page shows brochure (PDF), logo, social links and admission information	P1	Done
CAT-6	Fee comparison using only published, source-referenced fee versions; totals from one fee calculator	P0	Done
CAT-7	Import official AISHE college/university lists (admin upload of .xlsx)	P1	Built, not yet imported

5.4 Eligibility and recommendations

ID	Requirement	Priority	Status
ELG-1	Compare student marks with an offering's minimum qualification, minimum percentage and entrance exams. UG uses Class 12, diploma uses Class 10, PG uses graduation	P0	Done
ELG-2	Result is one of eligible / potentially eligible (within 5 points, missing data or exam needed) / not eligible, always with reasons	P0	Done
ELG-3	Counsellor recommends offerings to a student; recommendations appear on the student's enquiry page	P1	Done
ELG-4	Admin-side recommendation screen	P2	API only, UI not built
ELG-5	Scholarship rules with preliminary eligibility check and student applications	P1	Done

5.5 Enquiry CRM

ID	Requirement	Priority	Status
ENQ-1	Student submits an enquiry (college, course, marks snapshot, preferences) and gets a unique reference; duplicate submissions are prevented	P0	Done
ENQ-2	Status pipeline with rules, reasons, evidence gates and append-only history (section 4.4)	P0	Done

ENQ-3	Assign to a counsellor manually, or automatically to the active counsellor with the fewest open enquiries (setting, off by default)	P0	Done
ENQ-4	Notes (internal or student-visible), contact attempts with outcome, follow-up dates, messages	P0	Done
ENQ-5	One-tap Call, WhatsApp (`wa.me/91…`) and Copy phone actions	P0	Done
ENQ-6	Document requests and secure student uploads (PDF/JPG/PNG, 5 MB, private)	P1	Done
ENQ-7	Counselling appointments with email reminder 24 hours before	P1	Done
ENQ-8	Admission applications tracked per enquiry	P1	Done
ENQ-9	Quick filters, search, callback requests, lead source tracking	P1	Done

5.6 Counsellor area

ID	Requirement	Priority	Status
CNS-1	Separate `/counsellor` area with its own menu: summary, My enquiries, settings	P0	Done
CNS-2	See and act on assigned enquiries only, including academics and eligibility	P0	Done
CNS-3	Daily email digest of follow-ups due today and overdue (after 09:00 IST, once per day)	P1	Done
CNS-4	Cannot reassign enquiries, confirm admissions or open any admin page	P0	Done

5.7 Admin area

ID	Requirement	Priority	Status
ADM-1	Dashboard: totals (students, active colleges, offerings, counsellors, enquiries), new/pending/converted/closed, charts, filters	P0	Done
ADM-2	College management: profile, statuses Active/Draft/Inactive, publish/unpublish, delete (unpublished and unused only), logo, brochure, gallery, social links, admission info, featured order	P0	Done
ADM-3	Course offerings per college: description, mode, minimum qualification, seats, eligibility, exams, deadline, image, brochure	P0	Done
ADM-4	Versioned fees: draft → publish (previous version superseded), source and verification date required	P0	Done
ADM-5	Counsellor management: create, edit, activate/deactivate (ends sessions), performance table (assigned, contacted, pending, converted, closed, overdue)	P0	Done
ADM-6	Enquiry reports by counsellor, college and course with date and status filters; CSV export (max 5,000 rows, formulas neutralised, audited)	P0	Done

ADM-7	Team: admins with permissions, invitation codes	P0	Done
ADM-8	Site settings (organisation, phone, WhatsApp, Instagram, Facebook, auto-assign), email templates, CMS pages with versions, announcements	P1	Done
ADM-9	Feedback and testimonials (published only with student consent and admin approval)	P1	Done
ADM-10	Audit log and email outbox views	P1	Done

5.8 Notifications and payments

ID	Requirement	Priority	Status
NTF-1	Transactional emails via Brevo: OTP, password reset, enquiry received/acknowledged/assigned, status changes, messages, document requests, appointments, reminders, counsellor invites, follow-up digests	P0	Done
NTF-2	Durable email queue in MongoDB with retries; internal notes never included in emails	P0	Done
PAY-1	Payment requests for the consultancy service charge via Razorpay, with server-side signature check and webhook	P2	Built, keys not configured

6. Non-functional requirements

6.1 Security

- Every permission is enforced on the backend (route guards and record ownership); the UI never decides access on its own.
- Strict Zod validation on every request; unknown fields and operator objects (e.g. `{"$ne": null}`) are rejected with 422, blocking NoSQL injection.
- Records belonging to someone else return 404, not 403, so their existence is not revealed.
- Security headers (helmet, CSP), CORS limited to the frontend origin, request body limits, rate limits on sign-in, OTP and sensitive routes.
- Invitation codes and OTPs stored only as HMAC hashes; secrets live in `.env` only, never in frontend code or logs.
- Uploads checked by file content (magic bytes) and size, stored privately, served with `no-store`, `nosniff` and sandbox headers.
- Every admin and counsellor write is recorded in the audit log.

6.2 Privacy

- Student-facing responses are built from a whitelist of fields; internal notes are never shown to students.
- Students can request correction or deletion; completing a deletion anonymises the student.
- Testimonials show first name and city only, and only with consent.

6.3 Data integrity and honesty

- No fabricated colleges, fees, images or URLs. Sample data is clearly labelled and removable (`npm run seed:remove-samples`).
- Directory states its coverage; imported institutions are labelled "Basic listing – courses and fees not yet verified" until an admin adds them.
- Database changes happen only through explicit migration scripts, never automatically at startup.

6.4 Usability, accessibility and devices

- Responsive layouts for phones and desktops; no sideways scrolling (checked by automated mobile tests on key pages).
- Accessible labels, keyboard navigation, visible focus states; loading skeletons and confirmation dialogs for destructive actions.

6.5 Performance and reliability

- Search filters use indexed, precomputed fields instead of joins; the college finder searches an in-memory cache.
- Email queue and reminders survive restarts (atomic claim and lock in MongoDB).
- Optimistic concurrency on status changes prevents two people overwriting each other.

6.6 Quality gates

Before any release: `npm run lint`, `typecheck`, `test` and `build` must pass, plus Playwright end-to-end tests on desktop and phone viewports. Current result: 168 backend and 77 frontend tests, 13 end-to-end and 10 mobile tests passing.

7. System architecture

```
Browser ──► Next.js frontend (:3000) ──fetch──► Express API (:5100, /api) ──► MongoDB
(education_consultancy)
                                                          │
                                                          ├──► Brevo (email)      ├──► Razorpay
(payments)
                                                          └──► Wikidata (college finder data,
weekly)
```

Layer	Technology	Responsibility
Frontend	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS v4	Pages, UI components, role-based areas (`/student` , `/counsellor` , `/admin`). No secrets.
Backend	Node.js, Express 5, TypeScript, Zod	REST API: routes → validation → controllers → services → models; one error handler with a consistent error shape
Database	MongoDB with Mongoose 8	Collections with indexes and timestamps; separate test and e2e databases
Background work	In-process workers	Email queue, appointment reminders (every 5 min), follow-up digests, Wikidata refresh
API docs	OpenAPI generated from code	Swagger UI at `/api/docs`
Deployment	Dockerfiles and compose (project name `education-consultancy`)	Isolated from all other applications on the machine

8. Data model

Model	Purpose
User	Accounts: role, permissions, verified email, status, two-step sign-in
Session, AuthToken, EmailOtp, InviteCode	Sign-in sessions, reset/set-password tokens, verification codes, staff invitation codes (all secrets hashed)
StudentProfile	Academic record (Class 10/12/graduation marks), exams, preferences
Location, Course	Region/state/city hierarchy; course catalogue
College	College profile, status, media, social links, admission info, derived search fields
CollegeCourse	Offerings: duration, seats, mode, eligibility, exams, deadline, media
CollegeFee	Versioned fees (draft, published, superseded)

Enquiry, EnquiryEvent	Enquiry case with student snapshot; append-only history and conversation
AdmissionApplication, Appointment	Applications and counselling sessions
Scholarship, ScholarshipApplication	Scholarship rules and student applications
Shortlist, SavedComparison	Saved colleges and fee comparisons
StoredFile, Feedback, DataRequest	Private uploads, feedback/testimonials, privacy requests
Payment, SiteSettings, ContentPage, Announcement, EmailTemplateOverride	Service-charge payments, site settings, editable pages, announcements, email templates
CollegeImportJob, DirectoryReviewItem	AISHE import jobs and items needing review
EmailDelivery, AuditLog, Counter	Email queue, audit trail, yearly reference numbers

9. External integrations

Service	Use	Status
Brevo	Transactional email (HTTP API or SMTP)	Configured
Wikidata (CC0)	Real-time college finder data, cached and refreshed weekly	Live
Wikimedia Commons	Licensed campus photos with credits	Live
WhatsApp (wa.me links)	Chat links for students, counsellors and admins	Live
AISHE (Ministry of Education)	Official college/university lists, imported by admin upload	Ready, not imported
Razorpay	Service-charge payments	Keys needed

10. Success metrics

Metric	Definition	Where measured
Enquiry volume	New enquiries per day/week, by lead source	Admin dashboard
First-contact time	Time from NEW to CONTACTED	Enquiry history
Conversion rate	ADMITTED ÷ all enquiries in the period	Enquiry reports
Counsellor performance	Assigned, contacted, pending, converted, closed, overdue follow-ups per counsellor	Counsellors page
Overdue follow-ups	Open enquiries whose follow-up date has passed	Counsellor and admin views
Profile completion	Share of students with marks entered (needed for eligibility)	Reports
Catalogue quality	Published colleges, active offerings, colleges with verified fees	Dashboard, directory coverage
Student satisfaction	Average feedback rating	Feedback

11. Acceptance criteria

Criterion	Status
A student registers, verifies email by OTP and logs in	Done
Counsellor and admin can sign up only with a valid invitation code and land on their own dashboard	Done
A student completes the academic profile and sees eligibility on recommendations	Done
A student searches colleges in real time by name, location and branch and opens the official website	Done
A student submits an enquiry and receives a reference number; it appears in the admin dashboard	Done
An admin assigns the enquiry; the counsellor sees only their own enquiries and can call/WhatsApp, note, recommend and update status	Done
A student cannot access another student's records; a counsellor cannot access admin resources	Done
Admins manage colleges, courses, fees, media, scholarships, feedback and counsellors	Done
Reports by counsellor, college and course; CSV export	Done
Emails are delivered through the configured provider	Done

Works on mobile and desktop	Automated checks pass; real-device check pending
Lint, type checks, tests, builds and end-to-end tests pass	Done

12. Release status, known gaps and risks

Item	Impact	Action before launch
Development database has no published colleges	Verified directory and course browser are empty	Admins add verified colleges, courses and fees; import AISHE lists
Wikidata finder coverage gaps (e.g. some states under-represented)	Some colleges not found by search	Supplement with AISHE import; show coverage honestly
Admin-side recommendation screen not built	Admins use the API or counsellor screen	Build in next release (P2)
Razorpay keys not set	Online payment disabled	Add keys if online payment is wanted
Privacy policy and terms use built-in text	Legal content not business-specific	Replace with the business's own text
Rate limits and workers run in a single API process	Several API instances would not share limits	Use a shared store (e.g. Redis) when scaling out
Docker images not built and run yet	Deployment path untested	Build and run containers on the target server
Development links use a LAN IP	Email links work only on the same network	Set `FRONTEND_URL` to the public domain with HTTPS
No real-device accessibility check	Possible small-screen or screen-reader issues	Manual test on phones and with a screen reader

13. Future scope

- Admin recommendation screen and recommendation analytics.
- SMS / WhatsApp Business API notifications (beyond `wa.me` links).
- Native or installable (PWA) mobile app.
- Counsellor calendar sync and online counselling sessions.
- Bulk import of verified fees from college prospectuses.
- Multi-branch support for consultancies with several offices.
- Hindi and regional-language interface.